

Predicting Suicide Deaths in Shandong Province, China.

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Introduction

Suicide is the act of voluntarily taking one's own life.³ Approximately 700,000 people worldwide die each year due to suicide.³ The People's Republic of China has one of the lowest suicide rates in the world.¹ However, they are one of the few countries that has a higher female suicide rate than males.¹ This rate is also higher in rural areas of China versus urban settings.¹ The motivation for this study is to investigate predictors of suicide (our response) with a particular interest in sex and location of residence. With this motivation in mind, the main goals/questions to be addressed in the study are:

1. Does sex and location of residence, among other factors (age, highest education level, and method of attempt) predict suicide deaths?
2. Is there an interaction between sex and location of residence?

By determining these possible predictors and interactions, public health officials can work to reduce the rate of female suicides in China.

Methods

The data was obtained from Kaggle and R was used to analyze said data. The data appears to have been collected in the Shandong Province of China, which is a rural community just east of the Taihang mountains (northeast China.)² Upon initial inspection, the data was found to be extremely clean. There were no missing values or errors in the data. The only changes made to the data set were some small typos corrected in the column names (method instead of

Method.) There were no overall formatting issues other than the data being almost completely categorical and not numeric, which required some extra steps in the analysis and coding.

The sample size was 2571, with one outcome (suicide death) and five predictors of death by suicide: location of residence (rural), age, sex, method of attempt, and highest education level. A table is shown below that breaks down the variables and their categories:

Variable Breakdown		
Variable	Type	Categories
Suicide Death (Outcome)	Binary (meaning 2 possible outcomes)	Died, Survived
Location of Residence (Rural) (Predictor)	Categorical (3 options)	Yes (Urban), No (Rural), Unknown
Age (Predictor)	Numeric	N/A
Sex (Predictor)	Categorical (2 options)	Female, Male
Method of Attempt (Predictor)	Categorical (8 options)	Cutting, Hanging, Jumping, Other, Other poison, Pesticide, Poison with type unspecified, Unspecified
Highest Education Level (Predictor)	Categorical (5 options)	Illiterate, Primary, Secondary, Tertiary, unknown
<i>Reference levels: Location (Urban), Sex (Male), Method of Attempt (Unspecified), Highest Education Level (Unknown)</i>		

Table 1. Breakdown of variables in dataset.

As one can see from the table, the dataset is mostly categorical data with many levels, especially for method of attempt. As such, the modeling for this dataset will have to account for binary outcomes and categorical predictors. The theory and justification for this will be discussed below.

Results

Explanation of Model

As stated above, the outcome is binary and most of the predictors are categorical, meaning ordinary linear regression would not be appropriate. This is due to the follow assumptions needed to conduct ordinary linear regression:

- 1) Variables to be independent of each other.
- 2) Residuals (difference between the predicted value and the actual value for the outcome variable) are independent and follow a normal distribution.
- 3) Variance of residual terms is equal.
- 4) Normal distribution of the outcome variable (This was violated, as the binary outcome is not normally distributed.)
- 5) No correlation between variables.
- 6) No autocorrelation (correlation within the variable's data.)

Since these assumptions were violated in the data set, an alternative modeling technique was used: logistic regression. Logistic regression has the following assumptions, which allow for a binary outcome variable and categorical data:

- 1) Variables to be independent of each other.
- 2) Independent variables and log odds (a function of logistic regression) are linear.
- 3) No correlation between variables.
- 4) No significant outliers.
- 5) Large sample size.

This allowed us to create the following (simplified for readable) logistic model:

$$\ln\left(\frac{\text{probability of Suicide Death}}{1-\text{probability of Suicide Death}}\right) = \text{Location of Residence(Predictor)} + \text{Age(Predictor)} + \\ \text{Sex(Predictor)} + \text{Method of Attempt(Predictor)} + \text{Highest Education Level(Predictor)}$$

The confidence level used to test the significance of these predictors was chosen to be 0.05.

Model Validation

First, before moving further with the model to answer the client's questions, we need to validate that the model is correct and will predict what we need. To do this, we used stepwise regression (a method used for variable selection for a model) to determine if any predictors needed to be removed. We used both the Akaike information criterion (AIC) and Bayesian information criterion (BIC) methods, which gave the same results. No variables needed to be removed. We also ran a Hosmer-Lemeshow goodness-of-fit test, which determines if the proposed logistic regression model is a good fit for the binary outcome data. The p-value was large (0.57745) indicating a good fit.

Next, to check the validity of the model, various diagnostic plots were created. We wanted to check that the overall assumptions for the logistic model hold for our data. As the data is categorical, correlation between variables would not be an issue. The sample size is also large, so that was assumption holds as well. This leaves the following assumptions to be checked: 1) residuals are linear and 2) presence of outliers.

First, we checked the distribution of the residuals using a modified (for categorical data)

QQ plot:

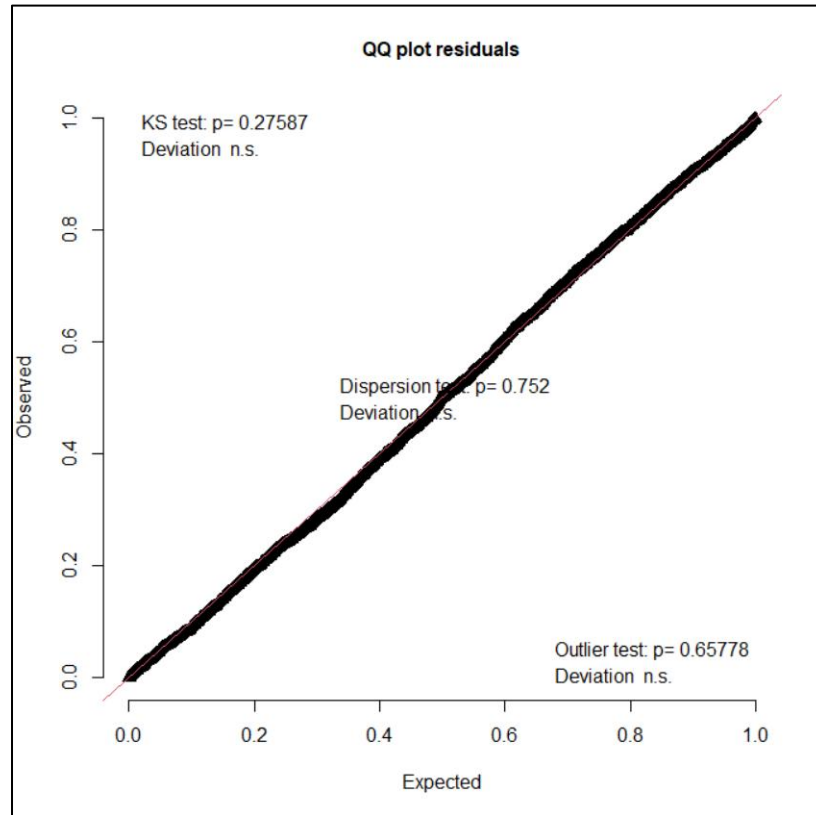


Figure 1. QQ Plot of Residuals.

In figure 1, it can be seen that the predicted and observed values for suicide death fall along the line when accounting for the categorical nature of the data. This indicates that the residuals are normally distributed. This graph also shows no outliers (outlier test showing a p-value greater than 0.05.) This was confirmed using a Cooks distance plot:

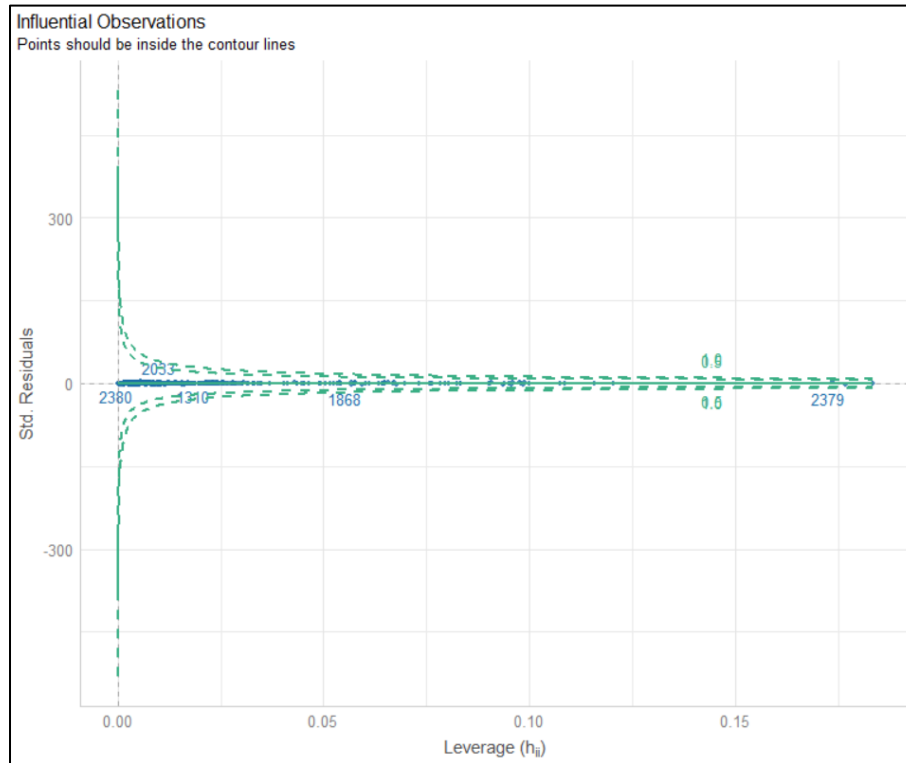


Figure 2. Cook's Distance Plot of Residuals.

As can be seen in figure 2, none of the points fall outside the dotted lines, which also indicates that there are no outliers.

Both assumptions hold for our data, but we also wanted to check the homogeneity (variance of the residuals is equal for all variables) of the residuals just to be safe:

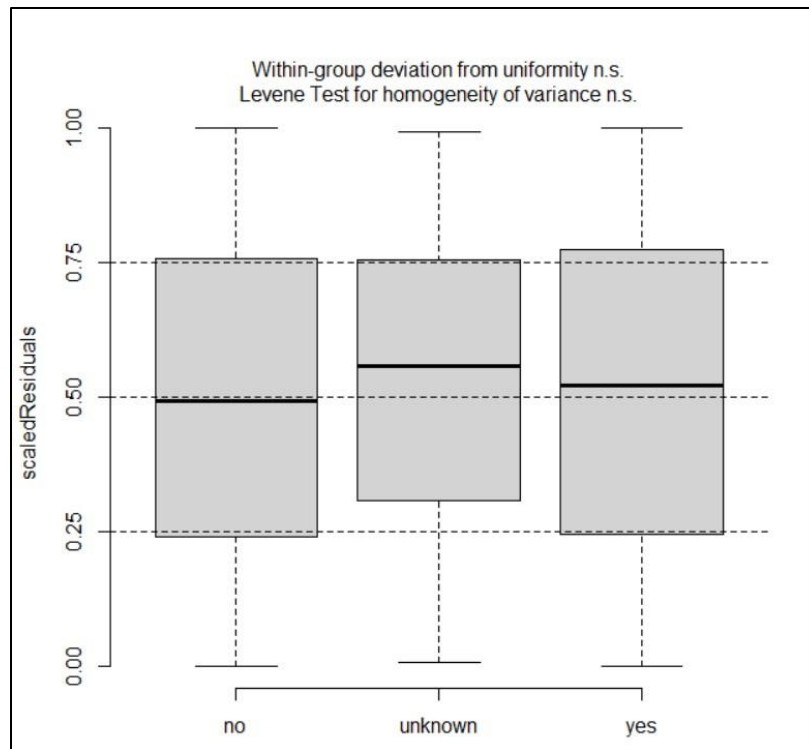


Figure 3. Homogeneity check for Location of Residence (Rural) variable (x-axis.)

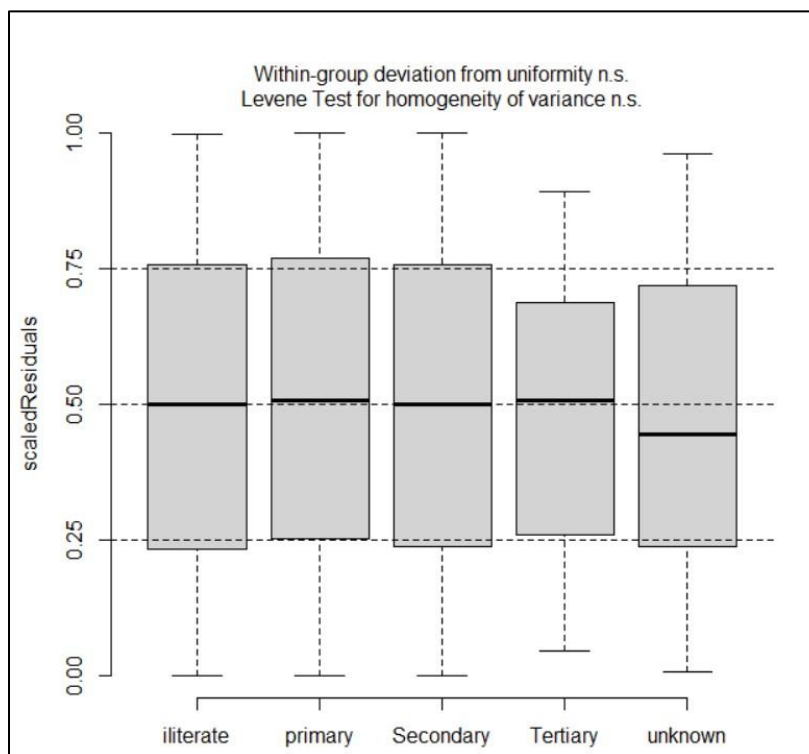


Figure 4. Homogeneity check for Highest Education Level variable (x-axis.)

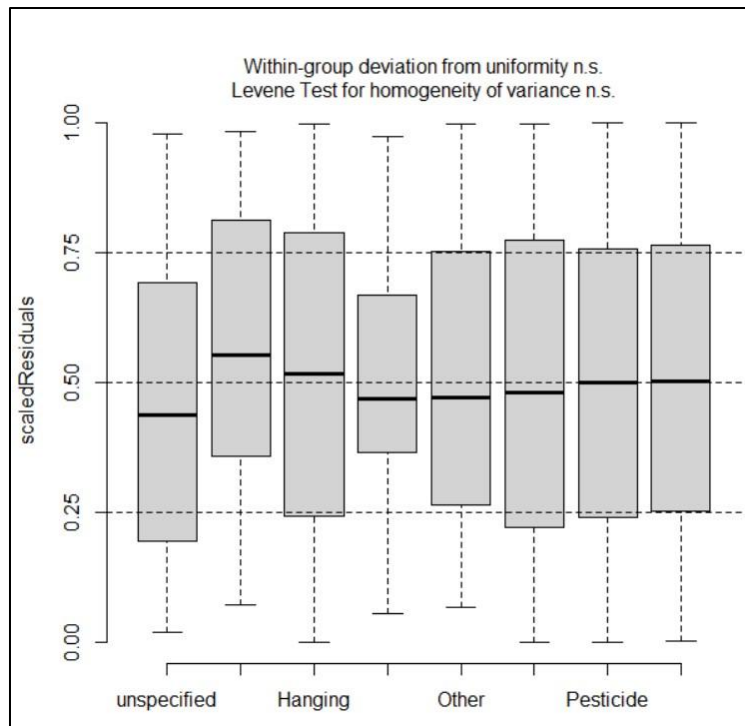


Figure 5. Homogeneity check for Method of Attempt variable (x-axis.)

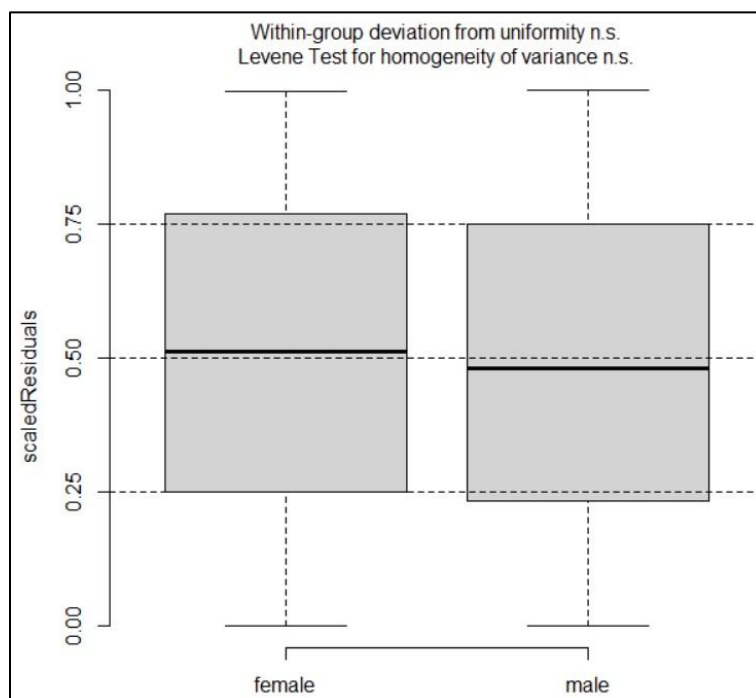


Figure 6. Homogeneity check for Sex variable (x-axis.)

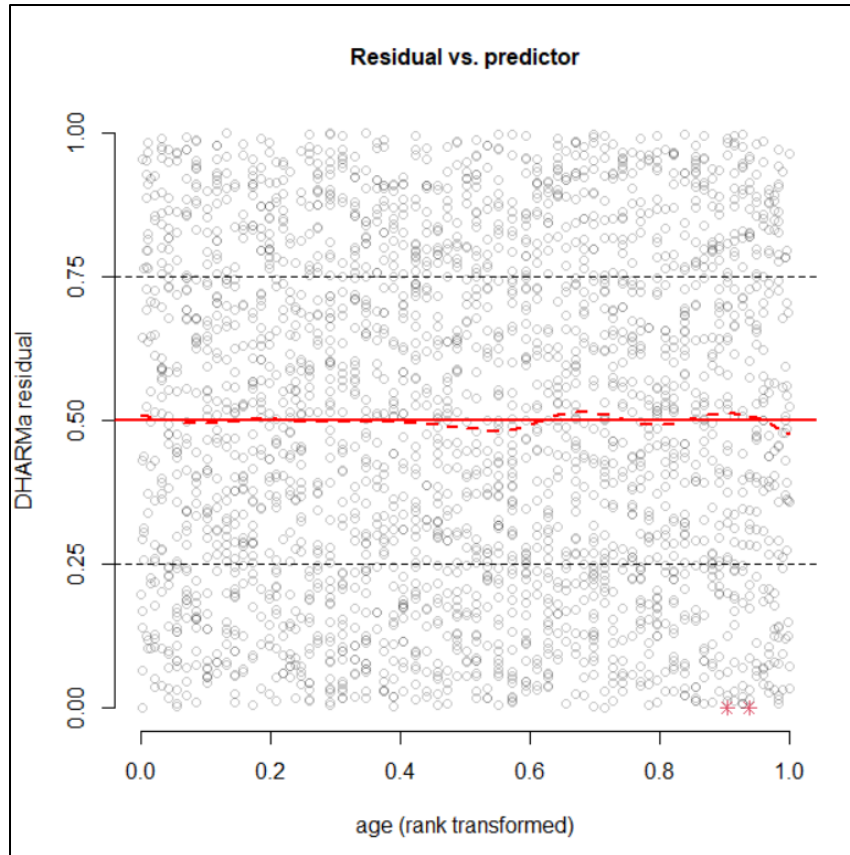


Figure 7. Residuals vs predicted values (homogeneity check for numeric variable, Age.)

Figures 3-7 show that the homogeneity of all the residuals were within the median value (horizontal line at 0.5), which indicates that the variances are equal. As the distribution of the residuals appears to be normal, no transformation was needed.

Next, we checked the accuracy of the results using a receiver operating characteristic curve (ROC) and a confusion matrix. Both an ROC and a confusion matrix tell us how efficient the model is at predicting the outcome (suicide death) accurately.

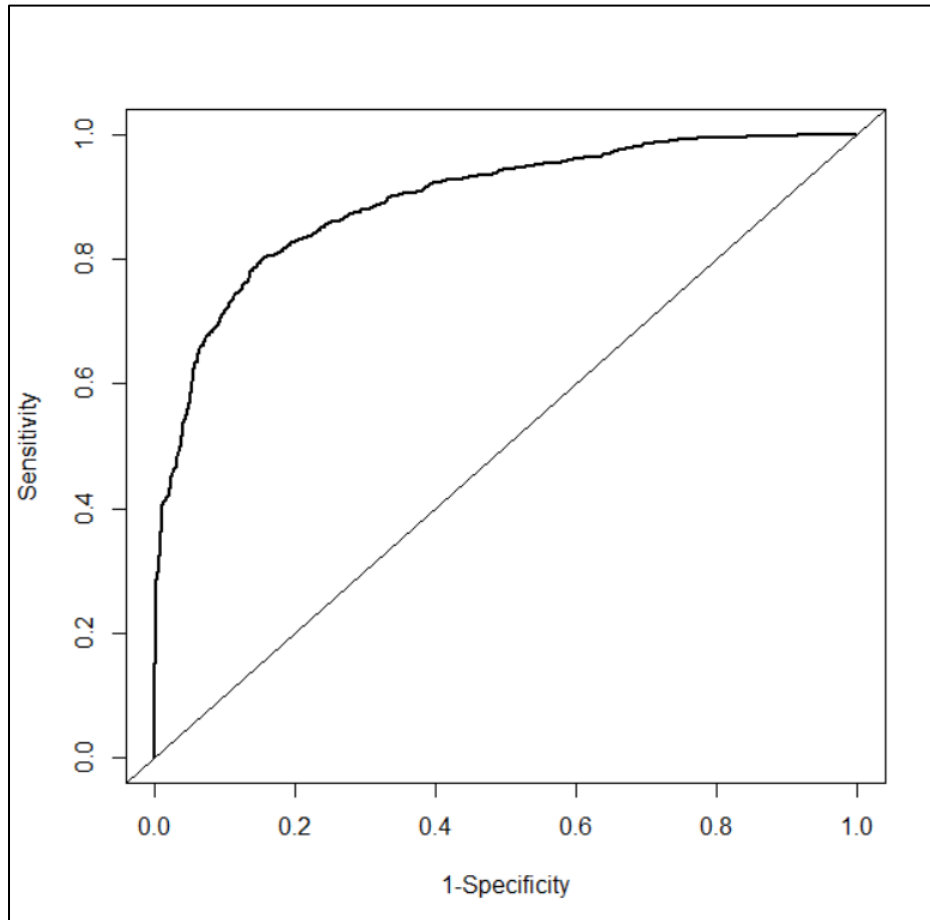


Figure 8. ROC for our logistic model.

In figure 8, we can see the probability of our model to correctly identify suicide death on the y-axis (sensitivity), while the x-axis has the probability that our model will incorrectly identify surviving a suicide attempt as suicide death (1-specificity, or the false positive rate.) The diagonal line displays what we would get from guessing or pure chance. The top left-hand corner (sensitivity of 1) represents perfect accuracy. From this graph, we can also get the area under the curve which is 0.8936 or 89%. This value tells us how well our model is at discerning between positive (died) and negative (survived) outcomes. The higher the value, the better. As our value is higher than 50%, the model is discerning greater than random chance.

Similarly to a ROC, we constructed a confusion matrix to determine the model's accuracy (probability that the model correctly predicts suicide death), sensitivity, specificity (the probability that the model will correctly identify surviving suicide death), positive predictive value (the probability the model will correctly identify suicide death when the individual truly passed) and negative predictive value (will correctly identify surviving suicide death when the individual truly survived.)

Confusion Matrix		
	Reference	
Prediction	0	1
0	1113	252
1	202	1004
Statistics		
Accuracy	0.8234	
Sensitivity	0.8464	
Specificity	0.7994	
Positive Predictive Value	0.8154	
Negative Predictive Value	0.8325	

Figure 9. Confusion matrix and statistics.

From figure 9, we found the following regarding the model:

- It correctly predicts suicide death 82% of the time.
- It correctly identifies suicide death 84% of the time.
- It correctly identifies surviving suicide death 79% of the time.
- It correctly identifies suicide death when the individual truly passed 81% of the time.
- It correctly identifies surviving suicide death when the individual truly survived 83% of the time.

For our model, accuracy is an important metric. So, the fact that our model is predicting suicide death correctly 82% of the time, which is greater than random chance (50% of the time), is very good.

Effects of the Predictors on the Response

Now that we have tested that the model is valid, it is time to answer the client's questions. No other model was evaluated as this model is easy to interpret, sufficiently answers the client's questions of interest and predicts the outcome of interest. As such, to answer the client's questions, we looked at the effects of the predictors on the response. First, we addressed the first question:

1. Does sex and location of residence, among other factors (age, highest education level, and method of attempt) predict suicide deaths?

and obtained the following results for our model:

Prediction Table				
Predictor	Coefficients	P-Value	Significant?	95% Confidence Intervals
Location of Residence: Rural	3.961	~ 0	Yes	(2.545, 5.971)
Location of Residence: Unknown	4.004	~ 0	Yes	(2.551, 6.033)
Age	0.019	~ 0	Yes	(0.011, 0.0256)
Sex: Female	0.332	0.003	Yes	(0.116, 0.548)
Method of Attempt: Cutting	3.474	0.038	Yes	(0.157, 3.387)
Method of Attempt: Hanging	2.714	~ 0	Yes	(4.709, 7.543)
Method of Attempt: Jumping	1.164	~ 0	Yes	(3.872, 8.207)
Method of Attempt: Other	2.994	~ 0	Yes	(4.260, 9.431)
Method of Attempt: Other poison	1.667	0.748	No	(-1.043, 1.787)

Method of Attempt: Pesticide	5.964	~ 0	Yes	(1.485, 4.057)
Method of Attempt: Poison with type unspecified	5.793	0.408	No	(-2.506, 1.062)
Highest Education Level: Illiterate	6.312	~ 0	Yes	(2.583, 4.488)
Highest Education Level: Primary	0.225	~ 0	Yes	(1.850, 3.707)
Highest Education Level: Secondary	2.586	0.013	Yes	(0.311, 2.150)
Highest Education Level: Tertiary	-0.722	~ 0	Yes	(1.490, 4.572)
<i>Reference levels: Location (Urban), Sex (Male), Method of Attempt (Unspecified), Highest Education Level (Unknown)</i>				

Table 2. Predictor significance.

From table 2, we can look at the coefficients of the model which will tell us the effect of our predictors on the *probability* of suicide death. For example, holding all the other predictors constant, for every unit increase in the numeric predictor age, the log odds of suicide death increase by 0.019. For our categorical predictors, the interpretation would look like this: holding all the other predictors constant, females are more likely to commit suicide death than males (our reference group) as the coefficient is positive and not negative. In contrast, for highest education level, those with tertiary education (graduate level) are less likely than those with unknown education to commit suicide death (the coefficient is negative.)

We can also see that the predictors in our model are all significant in all categories except for two methods of attempt: other poison, and poison with type unspecified. In other words, a formal answer for our client's first question:

1. Yes, sex and location of residence, age, highest education level, and method of attempt (except for other poison and poison with type unspecified) do predict suicide deaths in our data.

Next, we addressed the second question:

2. Is there an interaction between sex and location of residence?

All possible interactions were checked, and none were significant. So, to answer the clients second question:

2. No, there is no interaction between sex and location of residence when predicting suicide death.

This means that while sex and location of residence may predict suicide risk individually, their combination (say, female that lives in a rural area) does not influence suicide death. Only being either female or living in a rural area (independently) would predict suicide.

Conclusion:

The goal of this analysis was to determine if suicide death, a binary outcome, could be predicted by location of residence (rural), age, sex, method of attempt, and highest education level, with sex and location of residence being the focus. We implemented various validation techniques to make sure the model was sufficient before proceeding with analysis. These methods included: variable selection, goodness-of-fit test, residual diagnostic plots, ROC and confusion matrix.

The major findings of the report were that overall, sex (female), location of residence (rural), age, highest education level and method of attempt (except for other poison, and poison with type unspecified) significantly predict suicide. In contrast, sex and location of residence did not have an interaction, meaning that only the predictors independently predict suicide death. This

was found for all combinations of predictors, meaning that all predictors independently predict suicide death and are not modified by a third predictor (an interaction.)

These findings will benefit people in a significant way. If we can understand what predicts suicide death, we can allocate resources to help prevent them for at risk groups (females in rural settings.) China has a unique problem (in comparison to other countries) that they wish to reduce, which is female suicide deaths. Initially, it was of concern that females in rural areas (an interaction) have a greater risk of suicide death. However, this was not found to be the case for the Shandong province. Nevertheless, female sex was a significant predictor, which matches with the overall findings from previous studies.¹

For the future, it would be helpful to analyze data from other provinces, as looking at one province which is primarily rural may have biased the results. It would be beneficial to repeat the data collection in provinces that have more urban areas, or a combination of both rural and urban. It would also be helpful to look at other variables that might be influencing suicide death, such as income level or previously diagnosed mental health issues. The possibilities for expanding this study are endless.

References

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